

SAI PRASATH S

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🔄 saiprasaths 🔗 Portfolio Website

📁 INTERNSHIP

AI Intern, Ford Motors Private Limited

Jul 2025 – May 2026
Chennai, India

- Built **LLM-powered systems** for manufacturing quality analysis, enabling automated issue classification, root cause analysis.
- Developed and deployed **GenAI APIs** using **FastAPI** to integrate AI capabilities into enterprise workflows.
- Optimized **BigQuery data pipelines** operating at GB–TB scale, reducing processing time by approximately **30%**.
- Designed a **RAG-based enterprise knowledge platform** over **2,000+ documents** using hybrid retrieval with **FAISS and BM25**.
- Improved retrieval efficiency, indexing performance, and system reliability through optimized vector indexing and pipeline enhancements.
- Collaborated on AI-driven automation solutions to improve manufacturing quality analysis, knowledge discovery, and decision support.

Project Intern, Clowak Innovations- VIT-TBI

Jan 2023 – Mar 2023
Vellore, India

- Developed a **React Native mobile application** for user interaction and real-time system monitoring.
- Integrated **Google Firebase** for authentication, login management, and cloud-based data storage.
- Connected and integrated electronic hardware components including **headlights, side stand, indicators, and sensors** with the instrumentation cluster.
- Collaborated on embedded and mobile integration workflows to improve vehicle instrumentation and monitoring capabilities.

🎓 EDUCATION

B.TECH. ELECTRONICS AND INSTRUMENTATION ENGINEERING,

Vellore Institute Of Technology CGPA : 8.17

2020 – 2024
Vellore, India

M.TECH. COMPUTER SCIENCE AND ENGINEERING,

Vellore Institute of Technology CGPA: 9.10

2024 – 2026
Chennai, India

🧠 SKILLS

Languages (Python, Java, SQL, JavaScript, MATLAB) | **Machine Learning/Deep Learning** (Tensorflow, Keras, Scikit-learn, CNN, ANN, NLP) | **GenAI/LLM** (LLMs, RAG, Prompt Engineering, FAISS, pgvector, BM25, Hybrid Retrieval, Langchain) | **Backend** (FastAPI, Flask, Node.js, Express.js) | **Web Development** (HTML, CSS, JavaScript, React) | **Cloud & Databases** (GCP BigQuery, Google Firebase, AWS cloud)

📁 POSITION OF RESPONSIBILITIES

Instrument Society Of India - VIT, Projects Head

Jan 2022 – Jan 2023
Vellore, India

- Mentored **20+ students** in technical projects, prototyping, and implementation.
- Led project planning, coordination, and execution for student-led engineering initiatives.
- Organized and supported technical workshops to improve practical learning and hands-on development skills.

PROJECTS

CNN for Emotion classification in text, *(Python- NLTK, Pandas, Keras, Scikit-learn, Matplotlib)*

- Developed a **CNN-based NLP model** for multi-class emotion classification from text data.
- Built an **end-to-end machine learning pipeline** including text preprocessing, tokenization, stop-word removal, feature extraction, model training, and evaluation.
- Trained and optimized a **CNN architecture using Keras**, tuning hyperparameters to improve classification accuracy. Used **Scikit-learn** for model evaluation and performance analysis across multiple emotion classes.
- Improved multi-class prediction performance through preprocessing optimization and model tuning.

Mental Health Monitor (MHM) – Multi-Agent LLM System,

(Python, Flask, DistilBERT, LLMs, pgvector, NLP, Vector Embeddings, Chart.js, RBAC)

- Designed and implemented a **multi-agent LLM-driven mental health monitoring system** with context-aware conversational AI for continuous user interactions.
- Engineered a **Long-Context Cognitive Mapping memory layer** using **pgvector and vector embeddings** to retain longitudinal user context, achieving approximately **82% Top-K retrieval accuracy**.
- Built a **DistilBERT + LLM hybrid pipeline** for **4-class sentiment classification**, achieving approximately **88% emotion detection accuracy**.
- Developed a **MIND-SAFE crisis detection framework** to override unsafe LLM responses and trigger real-time human intervention alert workflows.
- Implemented a **real-time analytics dashboard** to monitor user sentiment trends, crisis indicators, conversation insights, and system performance metrics.
- Integrated **role-based access control** to protect sensitive user data, enforce privacy-aware access permissions, and support secure human-in-the-loop intervention.
- Optimized performance through **parallel Flask-based multi-agent orchestration**, reducing latency by approximately **30%** and achieving **sub-2-second response time** with approximately **95% reliability**.

HACKATHONS

DATASET' 24 Hackathon

- Participated in a data-centric hackathon involving analytical problem solving, dataset interpretation, and AI/ML-based solution development
- Explored data preprocessing, pattern analysis, and model-oriented approaches for extracting insights from structured and unstructured datasets under competitive timelines
- Strengthened practical experience in teamwork, technical presentation, and rapid experimentation with machine learning workflows

Daimler Hackathon

- Participated in an industry-focused hackathon centered on solving real-world automotive and industrial technology challenges through collaborative problem-solving and rapid prototyping
- Worked in a cross-functional team environment to analyze use cases, develop technical approaches, and present scalable solution ideas within constrained timelines
- Gained exposure to enterprise manufacturing workflows, industrial systems, and large-scale operational processes through technical sessions and plant interaction activities

CERTIFICATES

Full Stack Web Development - SHAPEAI  | Python - GUVI  | AWS Solution Architect Associate 

PUBLICATIONS

Real-time ECG Analysis and Classification using Neural Networks in IoT devices ,

IEEE i-PACT 2023 | DOI: 10.1109/i-PACT58649.2023.10434526

- Developed a **portable hardware setup** for real-time ECG signal acquisition and preprocessing, powered by smartphones or laptops.
- Implemented an **IoT-based data transmission pipeline** to send ECG signals to a dedicated system for analysis.
- Used a **pre-trained neural network model** for ECG classification and diagnostic prediction.
- Enabled diagnosis results to be transmitted back to smart devices for accessible and real-time health monitoring.